16/7/24

Basic Networking Commands.

Exp: 1

AIM:

To execute basic networking commands in Linux and windows operating system.

Networking Commands:

1. arp -a :-

It displays the ~~ARP~~ ARP Cache table (a (IP to -MA address mappings)

2. hostname:

Display the Computer's host name

3. ipconfig/all;

Shows Complete TCP/IP Configuration details.

4. nbtstat -a;

Displays Net Bios Information and troubleshoot details.

5. netstat:

Display active ~~Con~~ Connection., listing ports, routing ta and interface statistics Eg: netstat -r

6. nslookup:

It performs DNS queries and display DNS information.

Eg: nslookup www.google.com.

```
PS C:\Users\DELL> hostname
HDC8124162
PS C:\Users\DELL> ipconfig / all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                       (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.


The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is removed.

Examples:
    > ipconfig                      ... Show information
    > ipconfig /all                 ... Show detailed information
    > ipconfig /renew               ... renew all adapters
    > ipconfig /renew EL*           ... renew any connection that has its
                                        name starting with EL
    > ipconfig /release *Con*       ... release all matching connections,
                                        eg. "Wired Ethernet Connection 1" or
                                            "Wired Ethernet Connection 2"
    > ipconfig /allcompartments     ... Show information about all
                                        compartments
    > ipconfig /allcompartments /all ... Show detailed information about all
                                        compartments
PS C:\Users\DELL>
```

```
PS C:\Users\DELL> netstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\DELL> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:52956        HDC8124162:52956       ESTABLISHED
  TCP    127.0.0.1:52956        HDC8124162:52955       ESTABLISHED
  TCP    127.0.0.1:52957        HDC8124162:52958       ESTABLISHED
  TCP    127.0.0.1:52958        HDC8124162:52957       ESTABLISHED
  TCP    172.16.75.102:55921    hkg12s32-in-f10:https  TIME_WAIT
  TCP    172.16.75.102:56167    lcmaaa-ax-in-f10:https  TIME_WAIT
  TCP    172.16.75.102:58804    13.69.179.12:https     ESTABLISHED
  TCP    172.16.75.102:58807    52.123.226.20:https    ESTABLISHED
  TCP    172.16.75.102:58813    lcbwmp-in-f189:5228    ESTABLISHED
  TCP    172.16.75.102:60807    4.213.25.242:https     ESTABLISHED
  TCP    172.16.75.102:61755    pnmaaa-aq-in-f5:https  TIME_WAIT
  TCP    172.16.75.102:64829    cg-0s04-in-f5:https    ESTABLISHED
PS C:\Users\DELL>
```

Path ping :

It combines ping and traceroute functionality with statistics.

Ping :

(Packet ~~INTER~~ Internet Groper) Command is the best way to test the connection node. ping use ICMP (Internet Connection message protocol) to communication node.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping www.facebook.com)

Route:

route commands is used to show / manipulate the IP routing table It is to ~~set~~ setup static routes to specific host or networks via an interface.

Linux Commands;

ip : A versatile command for managing network interfaces IP addresses and routing tables. It is used to assign or deleted IPs tagged interface status.

- ip addr show
- ip addr del 192.168.1.254/24 dev emp3s
- ip link set eth0 promisc on
- ip route add default via 192.16.1.254 dev eth0



* ip link set eth0 down
* ip route get 10.10.1.4
* ip link set eth0 promisc on.

ipconfig:

A staple tool historically used for configuring and ~~fixing~~ troubleshooting network interfaces. It has large been replaced by the ip command.

mtr:

Combines the functionality of Ping and traceroute to provid ~~diag~~ diagnostic statistics for each network hop.

* mtr google.com
* mtr -g google.com
* mtr -b google.com
* mtr -c 10 google.com.

tcpdump. -D

- It is designed for capturing and displaying Packe

* dnf install -y tcpdump
* tcpdump -D
* tcpdump -i eth0
* tcpdump -i eth0 -n
* tcpdump -i eth0 -c host 8.8.8.8
* tcpdump -i eth0 net 10.1.10.124
* tcpdump -i eth0 port 53
* tcpdump -i eth0 port not 53 and not 25

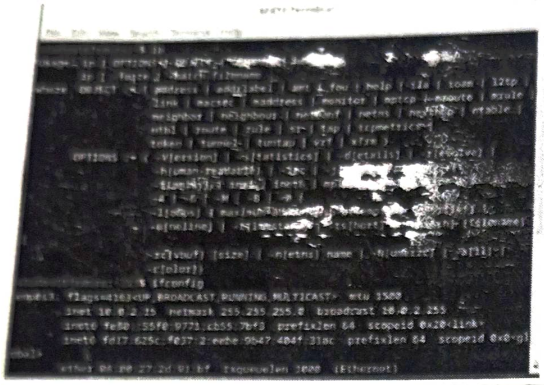
5. Ping :

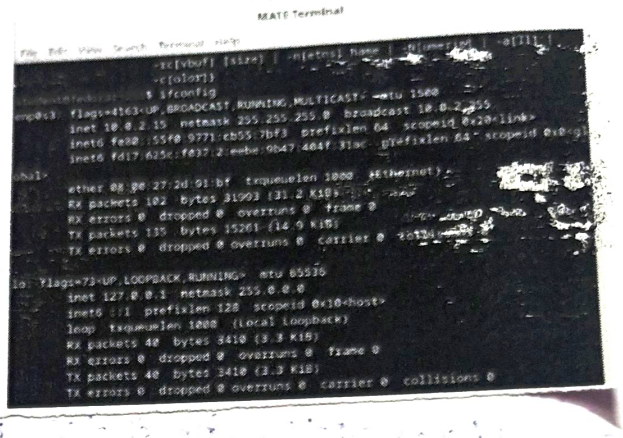
Verifies Ip-level Connectivity to a remort host by sending ICMP ~~eb~~ echo request messages It is the Primary tool for troubleshooting reachability network latency and name resolution.

* Ping google.com
* Ping -C 10 google.com.

Result:

Thus the networking commands for Windows and Linux operation Systems.




MATE Terminal
$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::55f0:9771:cb55:7bf3 prefixlen 64 scopeid 0x20<link>
inet6 fd17:625c:f037:2:ebe:9b47:404f:31ac prefixlen 64 scopeid 0x0<g
ether 08:00:27:2d:01:bf txqueuelen 1000 (Ethernet)
RX packets 102 bytes 31993 (31.2 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 135 bytes 15261 (14.9 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 40 bytes 3410 (3.3 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 40 bytes 3410 (3.3 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0